

AI Customer Support Bot

System Architecture Diagram — Interactive Reference

User Interface Layer

Layer 1

Where users interact with the AI bot across multiple platforms



User

Customer / End User



Chat Widget

Website Embedded UI



Mobile App

iOS / Android



Slack / WhatsApp

Messaging Platforms



Application Layer (Backend)

Layer 2

Backend logic that orchestrates all requests, auth, and routing



API Gateway

Rate Limiting & Routing



Authentication

OAuth / JWT / API Keys



Backend Server

FastAPI / Express



Session Manager

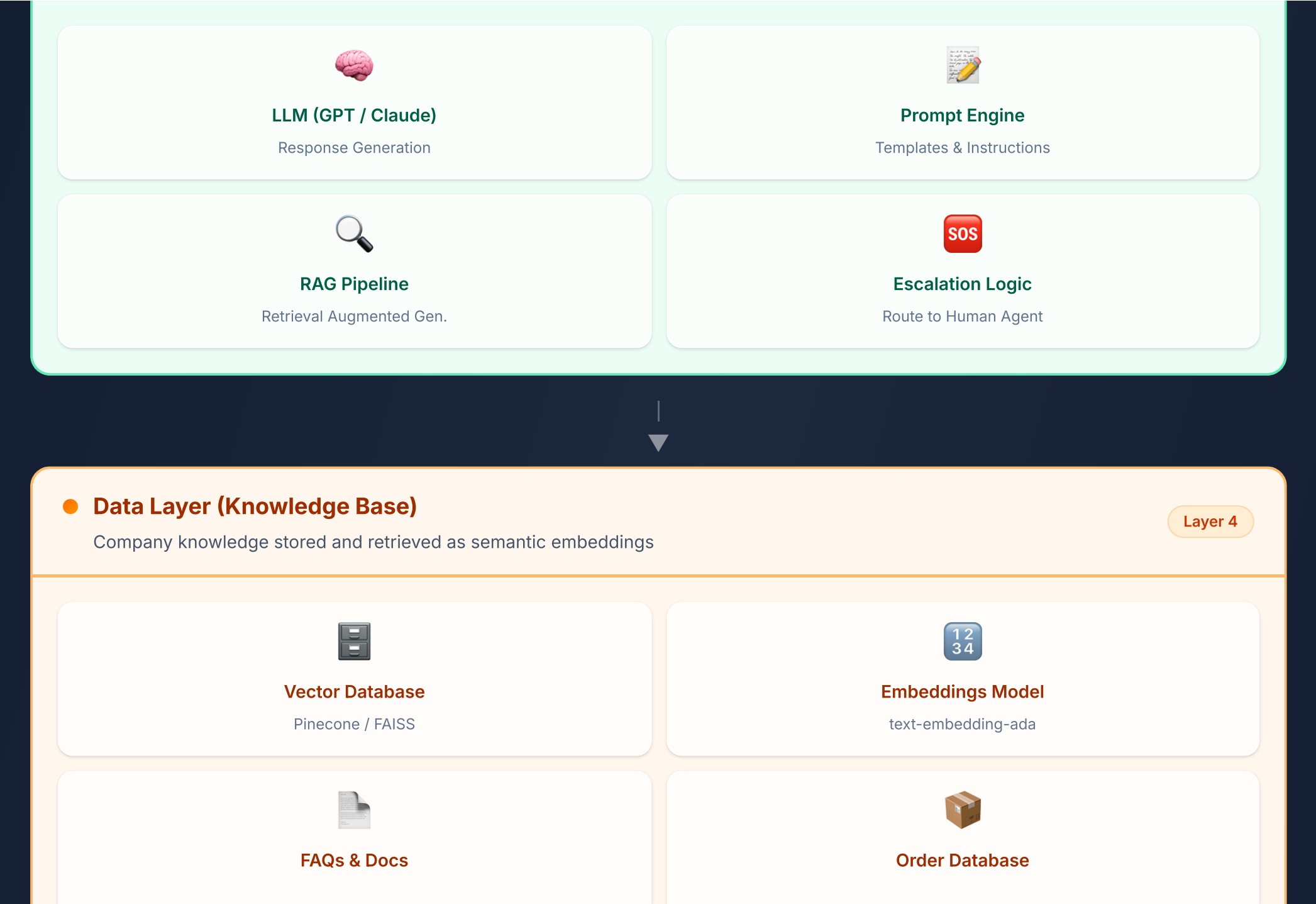
Conversation History



● AI / Model Layer

The intelligence engine — processes queries, retrieves context, generates answers

Layer 3



End-to-End Request Flow

Click a step to highlight it



User sends
question



API Gateway
validates



Backend routes
request



RAG retrieves
context



LLM generates
answer



Response
returned

Advanced Components

Cross-cutting concerns that support the whole system



Monitoring & Logging

Errors, latency, quality



Security Layer

Rate limits, input validation



Feedback Loop

Ratings improve the model over time

ARCHITECTURE SUMMARY

1. **User Interface** — Customer sends a message via chat, mobile, or messaging app

2. **Application Layer** — API gateway validates, authenticates, and routes the request
3. **AI / Model Layer** — RAG retrieves relevant docs; LLM generates a grounded response
4. **Data Layer** — Company FAQs, policies, and live data stored as vector embeddings

AI-Powered Customer Support Bot — System Architecture Diagram